COURSE: DATABASES

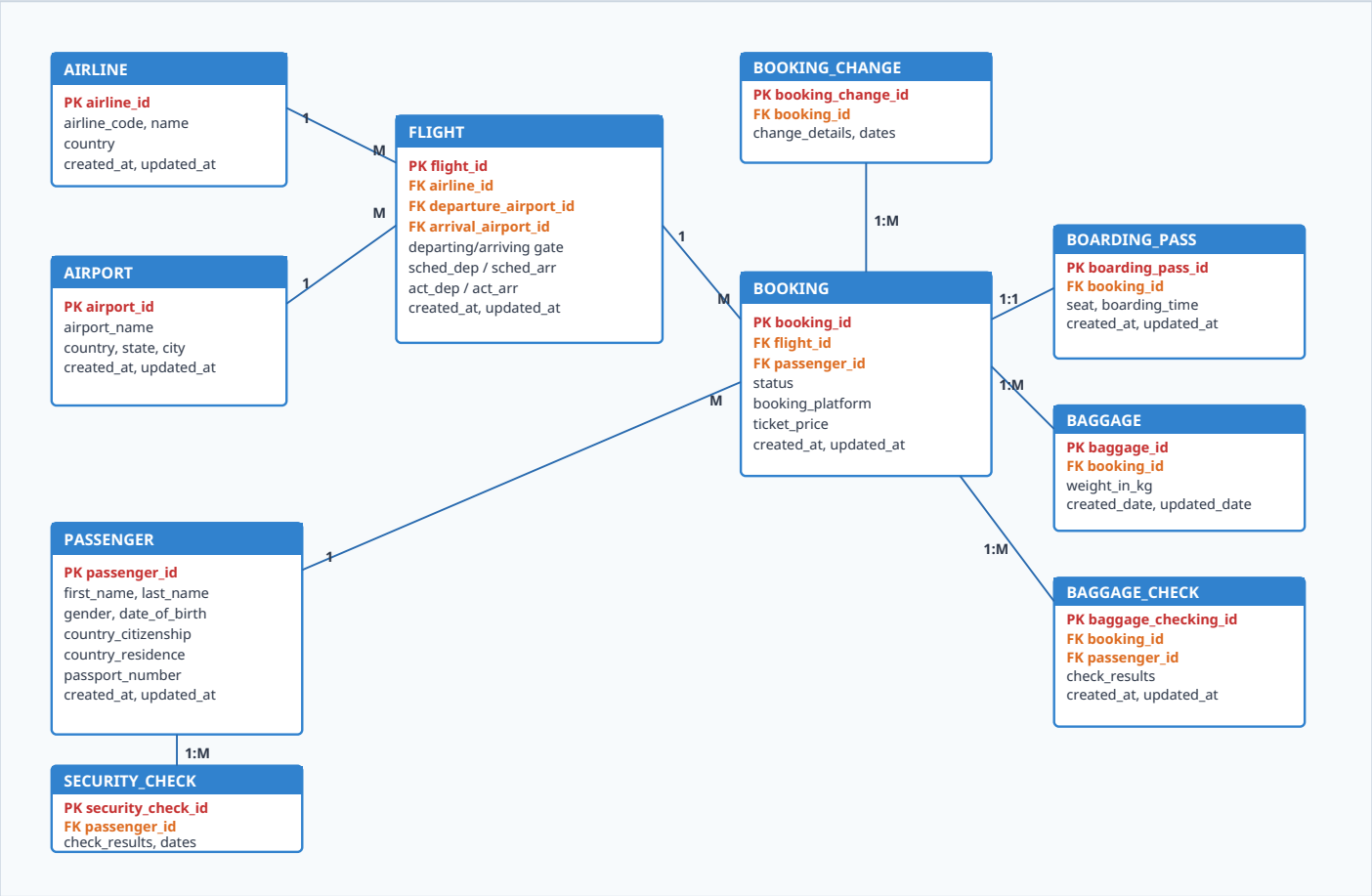
Laboratory Work 1: ERD Diagram & Solution

Domain: International Airport Database System

1. System Overview

This document contains the ER-Diagram and solution description for Laboratory Work 1 (Databases course). The system manages international airport operations: airlines, flights, passengers, bookings, boarding passes, baggage tracking, and security checks.

2. ER-Diagram (Relational Schema)



3. Entities & Attributes Description

AIRPORT			
Attribute	**Type**	**Key**	**Description**
airport_id	INT	PK	Unique ID of airport
airport_name	VARCHAR	-	Airport name
country / state / city	VARCHAR	-	Location fields
created_at / updated_at	TIMESTAMP	-	Audit timestamps

AIRLINE			
Attribute	**Type**	**Key**	**Description**
airline_id	INT	PK	Unique ID of airline
airline_code	VARCHAR	Unique	IATA/ICAO code
name / country	VARCHAR	-	Airline name and registration country

FLIGHT			
Attribute	**Type**	**Key**	**Description**
flight_id	INT	PK	Unique flight ID
airline_id	INT	FK	Links to AIRLINE
departure_airport_id	INT	FK	Links to AIRPORT (Departure)
arrival_airport_id	INT	FK	Links to AIRPORT (Arrival)
scheduled_departure / arrival	TIMESTAMP	-	Flight timetable

PASSENGER & BOOKING			
Entity	**Primary Key**	**Foreign Keys**	**Description**
PASSENGER	passenger_id	-	Stores personal info and passport data.
BOOKING	booking_id	flight_id, passenger_id	Flight ticket reservation records.
BOARDING_PASS	boarding_pass_id	booking_id	Issued seat and boarding pass data.
BAGGAGE	baggage_id	booking_id	Registered luggage records.
BAGGAGE_CHECK	baggage_checking_id	booking_id, passenger_id	Luggage safety check status.
SECURITY_CHECK	security_check_id	passenger_id	Passenger terminal security logs.

4. Relationships Summary

- **Airline to Flight (1:M):** One airline operates multiple scheduled flights.
- **Airport to Flight (1:M):** Airports host multiple departing and arriving flights.
- **Flight to Booking (1:M):** Each flight contains multiple ticket bookings.
- **Passenger to Booking (1:M):** A passenger can hold multiple flight bookings.
- **Booking to Boarding Pass (1:1):** Each valid booking produces one boarding pass upon check-in.
- **Booking to Baggage (1:M):** One booking can have several baggage items registered.